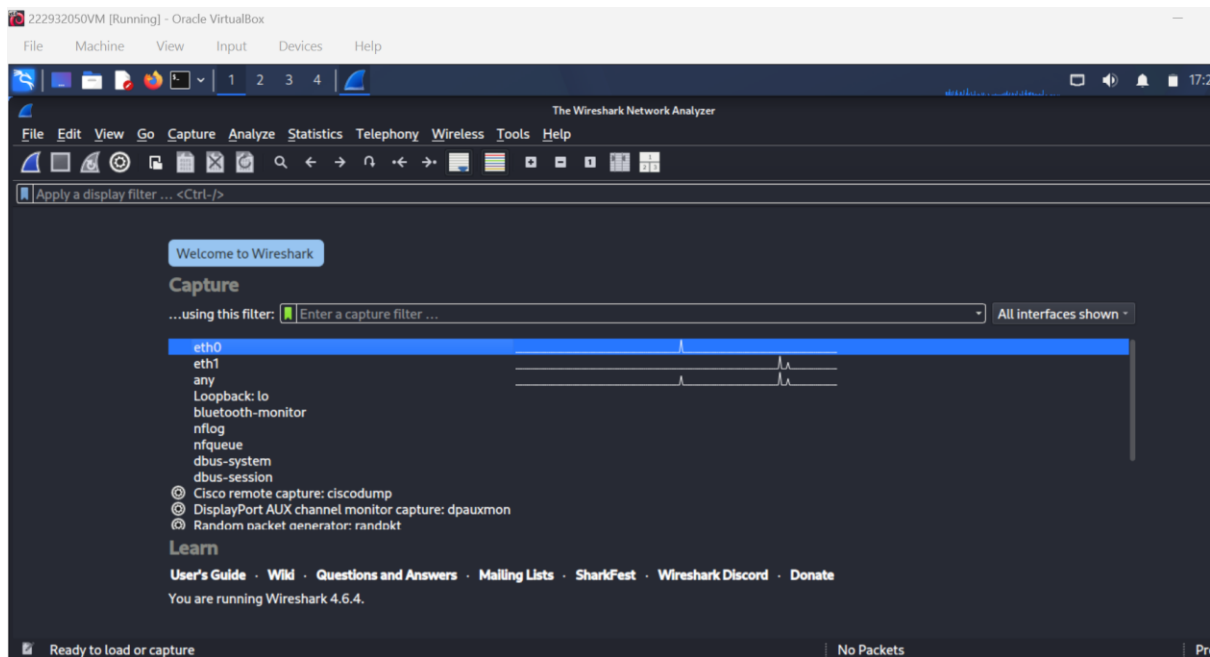


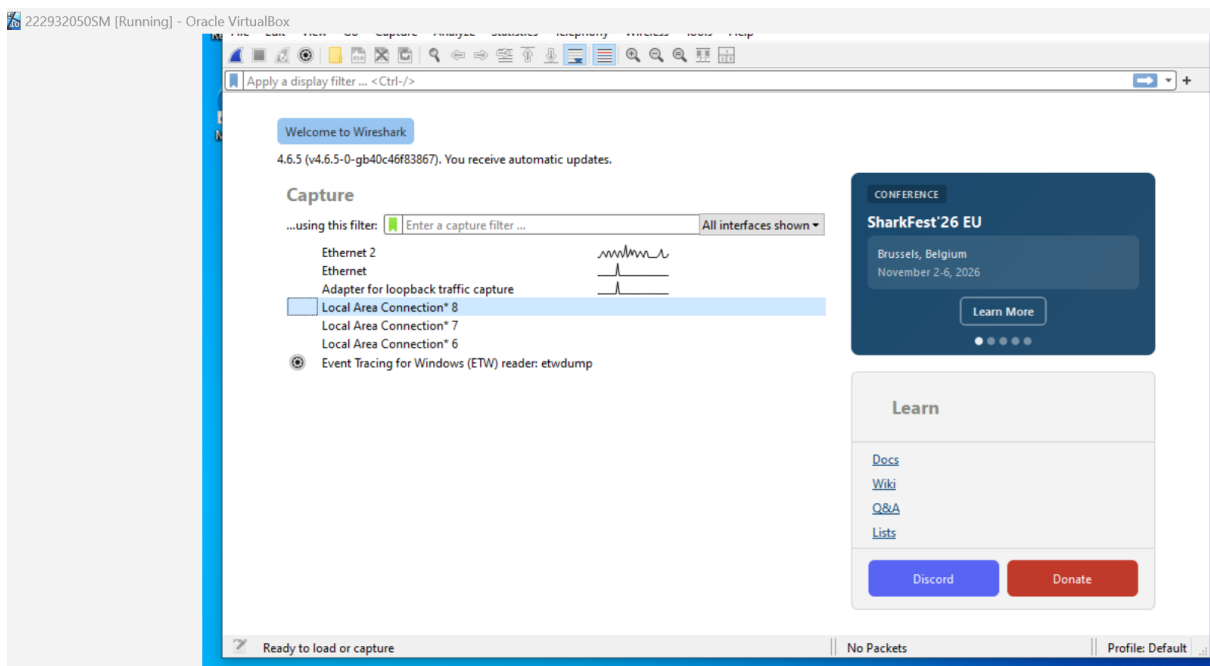
S MAHLANGU 222932050

LAB 8-10 PRACTICAL

LAB 1 : INSTALLING WIRESHARK ON KALI LINUX



LAB 2: Installing Wireshark on Windows Victim



LAB 3: Passive Sniffing using Wireshark

```
222932050SM [Running] - Oracle VirtualBox
Microsoft Windows [Version 10.0.19045.6456]
(c) Microsoft Corporation. All rights reserved.

C:\Users\222932050>ping 192.168.56.103

Pinging 192.168.56.103 with 32 bytes of data:
Reply from 192.168.56.103: bytes=32 time<1ms TTL=64
Reply from 192.168.56.103: bytes=32 time<1ms TTL=64
Reply from 192.168.56.103: bytes=32 time<1ms TTL=64
Reply from 192.168.56.103: bytes=32 time<1ms TTL=64

Ping statistics for 192.168.56.103:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\Users\222932050>
```

222932050VM [Running] - Oracle VirtualBox

File Machine View Input Devices Help

Capturing from eth1

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Apply a display filter ... <Ctrl-/>

No.	Time	Source	Destination	Protocol	Length	Info
16	290.230903221	PCSSystemtec_c4:24:...	Broadcast	ARP	60	Who has 192.168.56.103? Tell 192.168.56.101
17	290.230932937	PCSSystemtec_fc:e2:...	PCSSystemtec_c4:24:...	ARP	42	192.168.56.103 is at 08:00:27:fc:e2:85
18	290.231219023	192.168.56.101	192.168.56.103	ICMP	74	Echo (ping) request id=0x0001, seq=4/1024, ttl=128 (reply in 19)
19	290.231243341	192.168.56.103	192.168.56.101	ICMP	74	Echo (ping) reply id=0x0001, seq=4/1024, ttl=64 (request in 18)
20	293.356774150	PCSSystemtec_fc:e2:...	PCSSystemtec_30:83:...	ARP	42	Who has 192.168.56.1? Tell 192.168.56.103
21	293.357181206	PCSSystemtec_30:83:...	PCSSystemtec_fc:e2:...	ARP	60	192.168.56.1 is at 08:00:27:30:83:cf
22	294.259438053	192.168.56.101	192.168.56.103	ICMP	74	Echo (ping) request id=0x0001, seq=5/1280, ttl=128 (reply in 23)
23	294.259471439	192.168.56.103	192.168.56.101	ICMP	74	Echo (ping) reply id=0x0001, seq=5/1280, ttl=64 (request in 22)
24	295.404058440	PCSSystemtec_fc:e2:...	PCSSystemtec_c4:24:...	ARP	42	Who has 192.168.56.101? Tell 192.168.56.103
25	295.404574379	PCSSystemtec_c4:24:...	PCSSystemtec_fc:e2:...	ARP	60	192.168.56.101 is at 08:00:27:c4:24:07
26	296.926598805	192.168.56.101	192.168.56.103	ICMP	74	Echo (ping) request id=0x0001, seq=6/1536, ttl=128 (reply in 27)
27	296.926627915	192.168.56.103	192.168.56.101	ICMP	74	Echo (ping) reply id=0x0001, seq=6/1536, ttl=64 (request in 26)

Frame 1: Packet, 62 bytes on wire (496 bits), 62 bytes captured (496 bits) on 0
Ethernet II, Src: PCSSystemtec_fc:e2:85 (08:00:27:fc:e2:85), Dst: IPv6mcast_0010
Internet Protocol Version 6, Src: fe80::db5c:eb7b:bd96:7ea2, Dst: ff02::2
Internet Control Message Protocol v6

LAB 4: Password Sniffing Demonstration Using HTTP

LAB 5 : MAC FLOODING Demonstration

222932050VM [Running] - Oracle VirtualBox

File Machine View Input Devices Help

1 2 3 4

sm2229320

Session Actions Edit View Help

```
(sm222932050@kali)-[~]
$ macof -h
backdoor.sh: No such file or directory
Version: 2.5a2
Usage: macof [-s src] [-d dst] [-e tha] [-x sport] [-y dport]
        [-i interface] [-n times]

(sm222932050@kali)-[~]
$ sudo macof -i eth1 -n 10
d5:c9:c9:4b:53:ec ce:4b:69:1:2c:7b 0.0.0.0.15039 > 0.0.0.0.41941: S 846067547:84606754
4b:a5:7c:16:43:18 48:9b:6b:5e:c:51 0.0.0.0.44467 > 0.0.0.0.22626: S 1353235878:1353235
8e:96:14:f:a2:28 7:bc:fb:9:e2:ab 0.0.0.0.43205 > 0.0.0.0.5401: S 1729189894:1729189894
1b:de:ce:b:db:21 2a:98:42:4d:e9:29 0.0.0.0.5444 > 0.0.0.0.55239: S 311328303:311328303
5d:24:7d:7:7b:23 e3:24:27:6e:a6:bf 0.0.0.0.64992 > 0.0.0.0.8762: S 86459343:86459343(0
df:56:67:5f:5d:da 29:e4:8c:41:e7:12 0.0.0.0.36903 > 0.0.0.0.12037: S 1215121735:121512
dd:da:bc:76:62:27 9:a:af:4:7:73 0.0.0.0.19810 > 0.0.0.0.26255: S 1872592600:1872592600
2e:73:18:30:36:a3 c:ef:c3:16:11:98 0.0.0.0.33662 > 0.0.0.0.24764: S 1668056526:1668056
43:1c:da:54:ae:dc ab:a1:49:30:6d:0 0.0.0.0.13467 > 0.0.0.0.32700: S 1365726960:1365726
e9:67:cd:68:cd:31 44:a5:9b:59:f2:71 0.0.0.0.6403 > 0.0.0.0.62007: S 715087134:71508713

(sm222932050@kali)-[~]
$
```

222932050VM [Running] - Oracle VirtualBox

File Machine View Input Devices Help

Capturing from eth1

View Go Capture Analyze Statistics Telephony Wireless Tools Help

display filter ... <Ctrl-/>

Time	Source	Destination	Protocol	Length	Info
9 5.953391806	192.168.56.0	224.0.0.251	MDNS	119	Standa
9 5.953674722	fe80::1376:4222:f0c...	ff02::fb	MDNS	139	Standa
1 5.954059508	fe80::1376:4222:f0c...	ff02::1:3	LLMNR	95	Standa
2 5.954275695	192.168.56.0	224.0.0.252	LLMNR	75	Standa
3 6.119262129	192.168.56.0	224.0.0.22	IGMPv3	60	Member
4 6.119263409	fe80::1376:4222:f0c...	ff02::16	ICMPv6	90	Multic
5 48.657407723	192.168.56.103	192.168.56.1	DHCP	324	DHCP R
6 48.673785506	PCSSystemtec_30:83:...	Broadcast	ARP	60	Who ha
7 48.673834079	PCSSystemtec_fc:e2:...	PCSSystemtec_30:83:...	ARP	42	192.16
8 48.674574592	192.168.56.1	192.168.56.103	DHCP	590	DHCP A
9 53.888907089	PCSSystemtec_fc:e2:...	PCSSystemtec_30:83:...	ARP	42	Who ha
9 53.889768689	PCSSystemtec_30:83:...	PCSSystemtec_fc:e2:...	ARP	60	192.16
1 100.955771307	22.55.34.57	246.251.181.34	IPv4	54	
2 100.955970624	150.1.176.42	97.53.4.16	IPv4	54	
3 100.956072970	169.13.154.10	68.78.196.114	IPv4	54	
4 100.956191674	78.33.212.102	48.141.31.67	IPv4	54	
5 100.956311460	190.95.78.50	226.202.238.124	IPv4	54	

1: Packet, 85 bytes on wire (680 bits), 85 b 0000 01 00 5e 00 00 fb 0a 00 2

et II, Src: 0a:00:27:00:00:10 (0a:00:27:00:0 0010 00 47 39 b8 00 00 01 11 a

et Protocol Version 4, Src: 192.168.56.0, Ds 0020 00 fb 14 e9 14 e9 00 33 5

atagram Protocol, Src Port: 5353, Dst Port: 0030 00 00 00 00 00 00 0e 5f 6

ast Domain Name System (query) 0040 74 5f 6d 63 63 04 5f 74 6

LAB 6: DHCP Starvation Demonstration Using Yersinia

LAB7: ARP Poisoning Demonstration

```
Temporary IPv6 Address. . . . . : fd17:625c:f037:3:49b3:f6d7:b6e0:2e7
Link-local IPv6 Address . . . . . : fe80::d81e:b335:8a37:3a31%6
IPv4 Address. . . . . : 10.0.3.15
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : fe80::2%6
                             10.0.3.2
```

C:\Users\222932050>arp -a

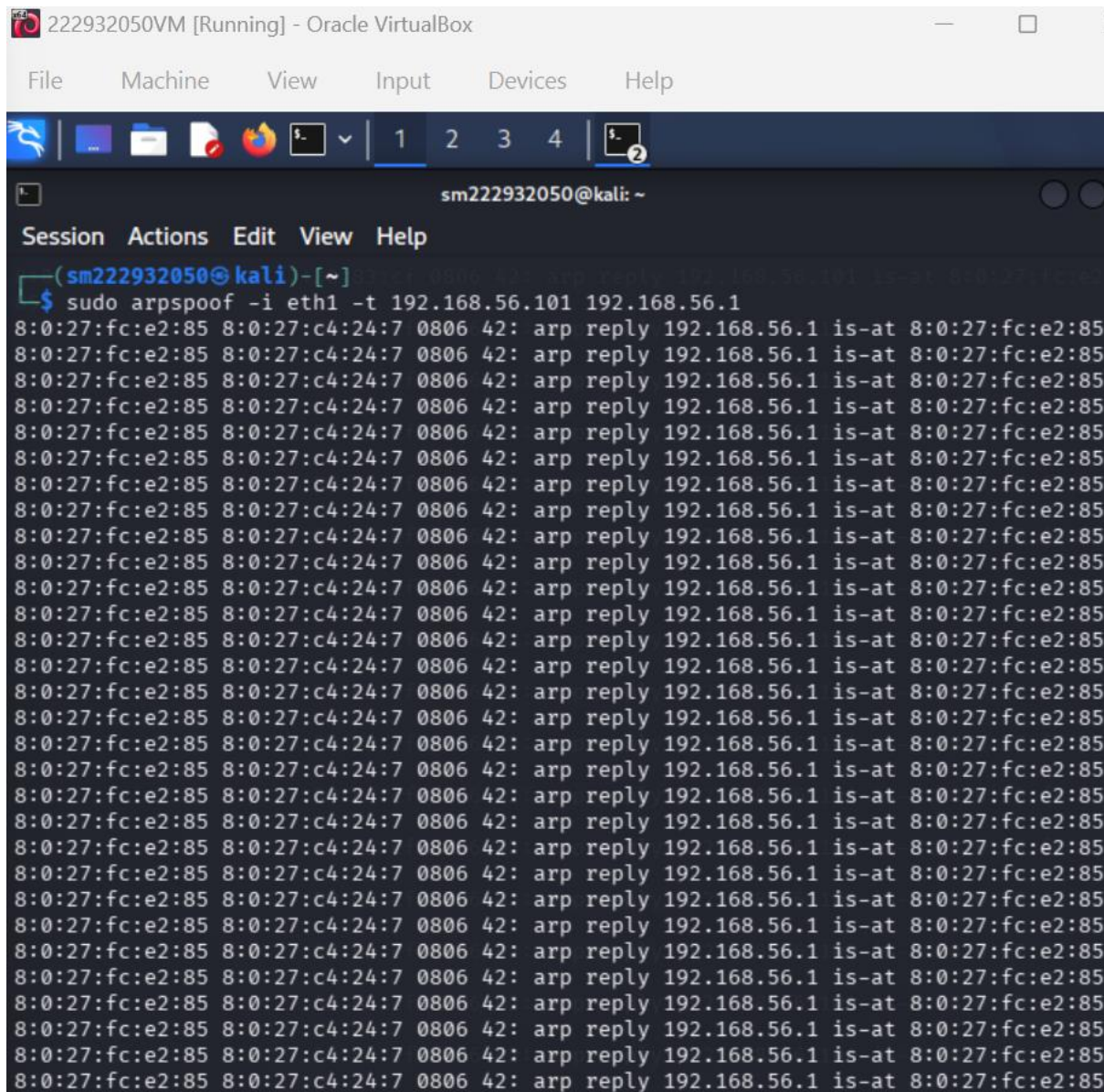
Interface: 10.0.3.15 --- 0x6

Internet Address	Physical Address	Type
10.0.3.2	52-55-0a-00-03-02	dynamic
10.0.3.255	ff-ff-ff-ff-ff-ff	static
224.0.0.22	01-00-5e-00-00-16	static
224.0.0.251	01-00-5e-00-00-fb	static
224.0.0.252	01-00-5e-00-00-fc	static
239.255.255.250	01-00-5e-7f-ff-fa	static
255.255.255.255	ff-ff-ff-ff-ff-ff	static

Interface: 192.168.56.101 --- 0x7

Internet Address	Physical Address	Type
192.168.56.1	08-00-27-fc-e2-85	dynamic
192.168.56.103	08-00-27-fc-e2-85	dynamic
192.168.56.255	ff-ff-ff-ff-ff-ff	static
224.0.0.22	01-00-5e-00-00-16	static
224.0.0.251	01-00-5e-00-00-fb	static
224.0.0.252	01-00-5e-00-00-fc	static
239.255.255.250	01-00-5e-7f-ff-fa	static

C:\Users\222932050>





222932050VM [Running] - Oracle VirtualBox

File Machine View Input Devices Help

Capturing from eth1

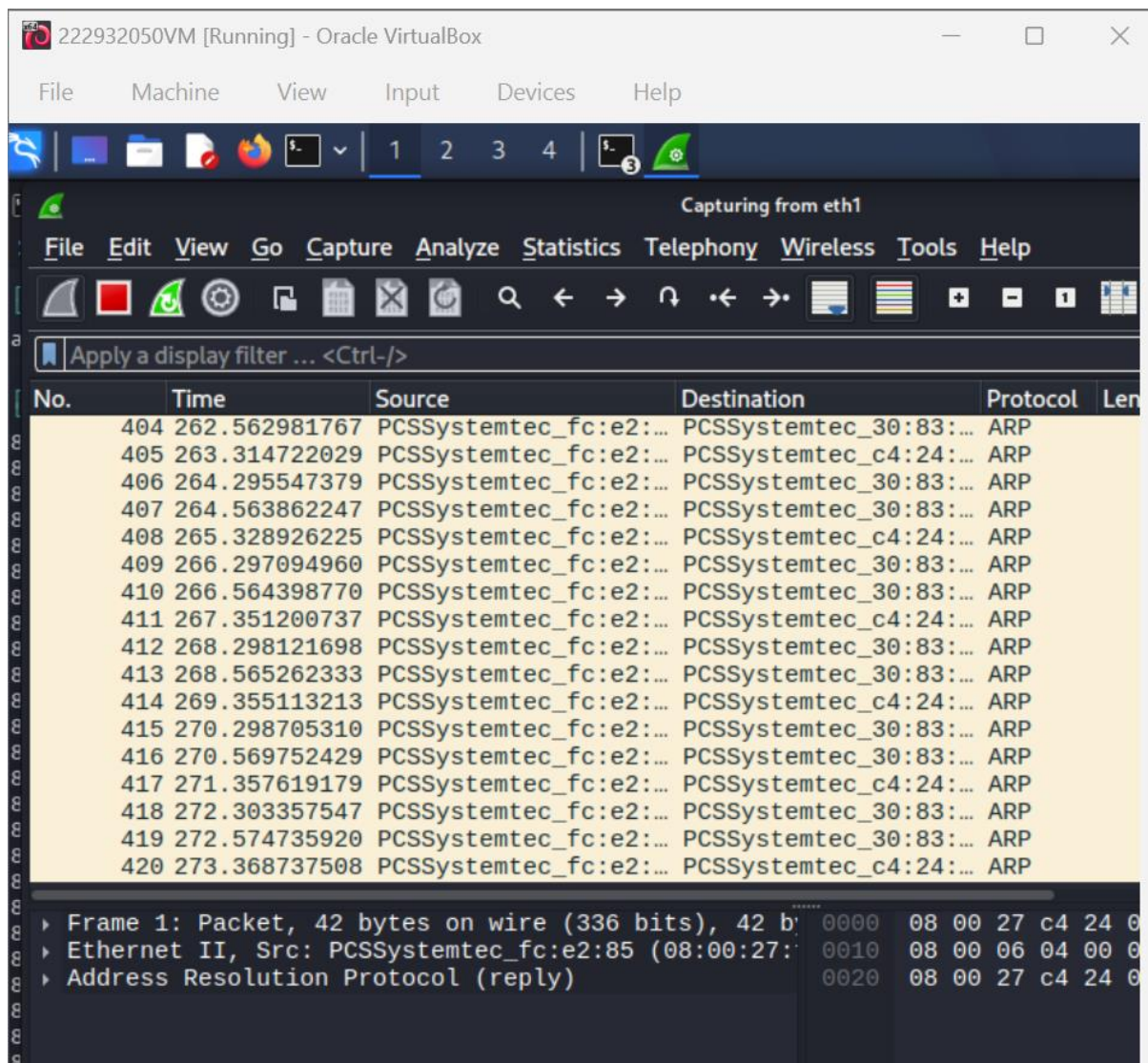
Session

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

ip.addr == 192.168.56.101

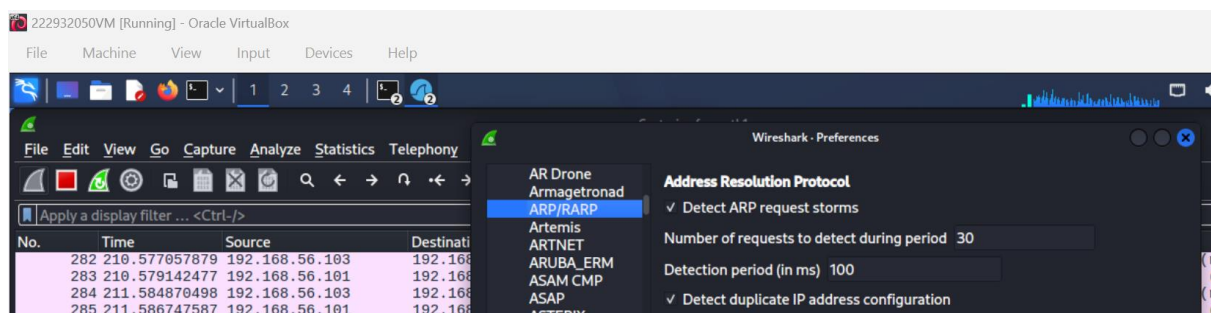
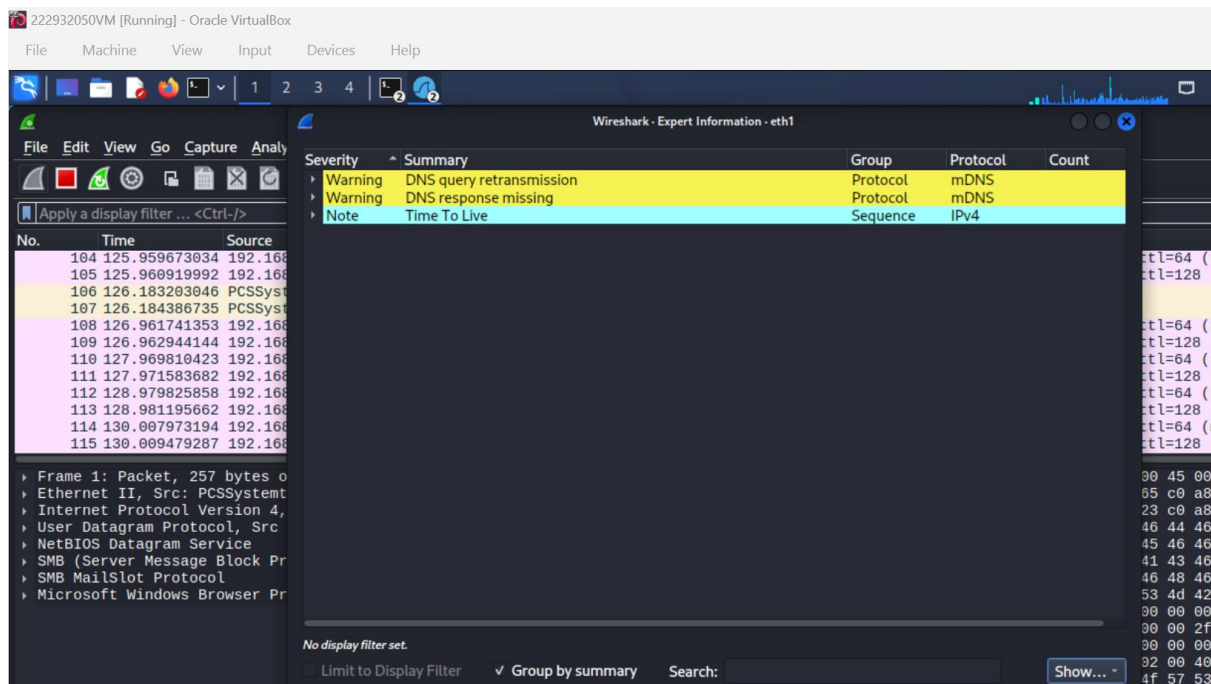
No.	Time	Source	Destination	Protocol
58	38.093214272	192.168.56.101	192.168.56.255	Broadcast

Frame 58: Packet, 243 bytes on wire (1944 bits), 200 bytes captured (1600 bits) on eth1
Ethernet II, Src: PCSSystemtec_c4:24:07 (08:00:27:0c:4c:07), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
Internet Protocol Version 4, Src: 192.168.56.101, Destination: 192.168.56.255
User Datagram Protocol, Src Port: 138, Dst Port: 138
NetBIOS Datagram Service
SMB (Server Message Block Protocol), Trans Request
SMB MailSlot Protocol
Microsoft Windows Browser Protocol

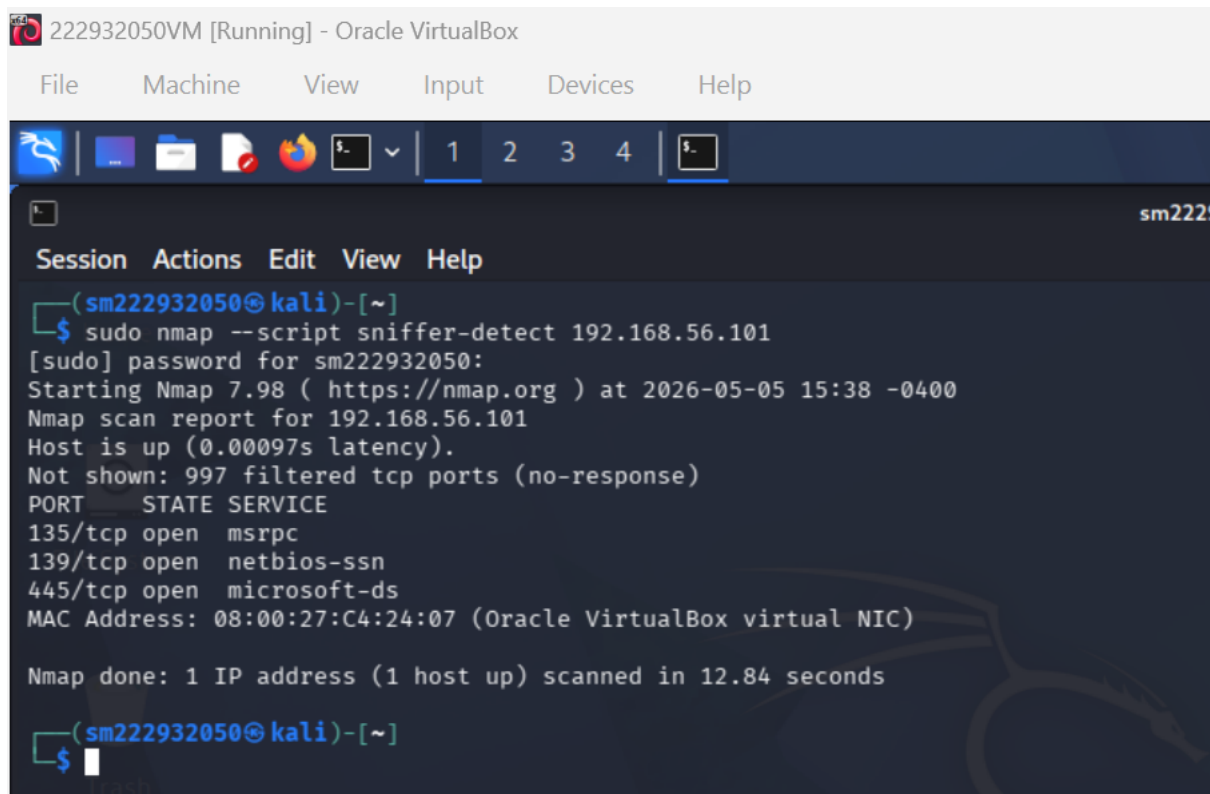


LAB 9: MAC Address Spoofing on Kali

LAB 10: Detecting ARP Poisoning in Wireshark



LAB 11: Detect Promiscuous Mode Using Nmap



```
(sm222932050@kali)-[~]
$ sudo nmap --script sniffer-detect 192.168.56.101
[sudo] password for sm222932050:
Starting Nmap 7.98 ( https://nmap.org ) at 2026-05-05 15:38 -0400
Nmap scan report for 192.168.56.101
Host is up (0.00097s latency).
Not shown: 997 filtered tcp ports (no-response)
PORT      STATE SERVICE
135/tcp   open  msrpc
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
MAC Address: 08:00:27:C4:24:07 (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 12.84 seconds
(sm222932050@kali)-[~]
$
```

LAB 12: Remote Packet Capture Using Wireshark

Passive sniffing is to monitor traffic without modifying it. Active sniffing is to manipulate switched networks. Organizations must enable port security to limit the number of mac addresses and Vlan segmentation in order to prevent sniffing attacks. This lab is only performed in an isolated VM environment to protect the main computer from being affected by these harmful simulation activities.

Chapter 9: Social Engineering Practical Lab


```
222932050VM [Running] - Oracle VirtualBox
File Machine View Input Devices Help

root@kali: /home/sm222932050

Session Actions Edit View Help

[---] The Social-Engineer Toolkit (SET) [---]
[---] Created by: David Kennedy (ReLiK) [---]
[---] Version: 8.0.3 [---]
[---] Codename: 'Maverick' [---]
[---] Follow us on Twitter: @TrustedSec [---]
[---] Follow me on Twitter: @HackingDave [---]
[---] Homepage: https://www.trustedsec.com [---]
Welcome to the Social-Engineer Toolkit (SET).
The one stop shop for all of your SE needs.

The Social-Engineer Toolkit is a product of TrustedSec.
Visit: https://www.trustedsec.com

It's easy to update using the PenTesters Framework! (PTF)
Visit https://github.com/trustedsec/ptf to update all your tools!

Select from the menu:
1) Social-Engineering Attacks
2) Penetration Testing (Fast-Track)
3) Third Party Modules
4) Update the Social-Engineer Toolkit
5) Update SET configuration
6) Help, Credits, and About
99) Exit the Social-Engineer Toolkit

set>
```

```
222932050VM [Running] - Oracle VirtualBox
File Machine View Input Devices Help

root@kali: /home/sm222932050

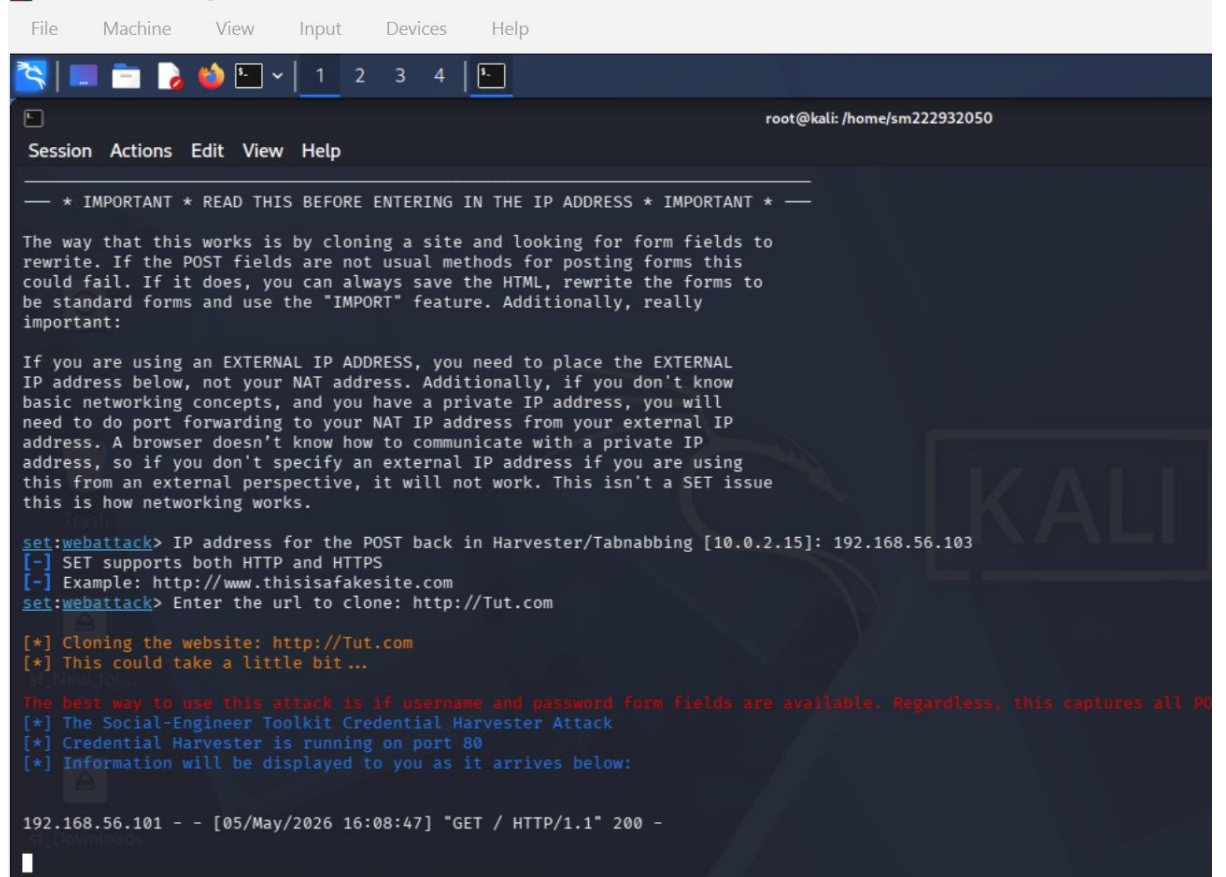
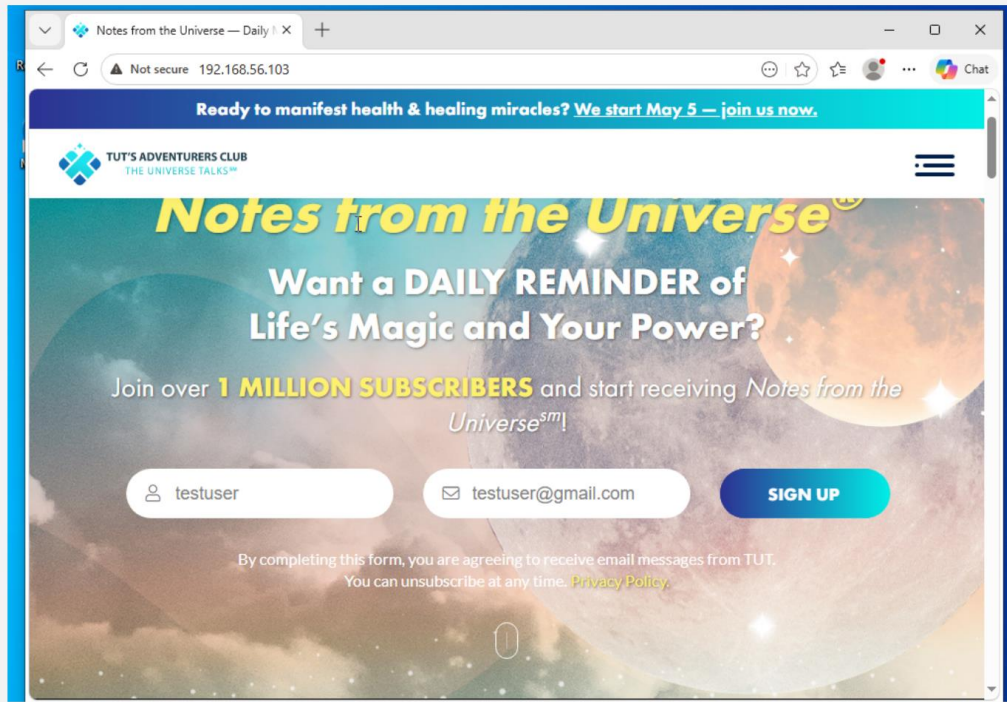
Session Actions Edit View Help

01011001011011110110100100000011100
10011001010110000101101100011011000111
10010010000001101000011000010111011001
10010100100000011101000110111100100000
011011010111010110001101101000001000
000111010001101001011011011001010010
000001101110110111000100000011100101
10111101101011100100010000001101000
01100001011011100110010001110011001000
0000111010001011010010100100100000101
0100011010000110000101101100110101101
1100110010000001100110011011110110010
001000000111010111001101101001011011
10011001110010000001110100011010000110
01010010000001010011011101110110001101
10100101100001011011000010110101000101
0110110011001110110100101110011001
010110010111001000100000010101000110
11110110111101101100011010110110100101
11010000100000001010100110100001110101
011001110111001100101010

[---] The Social-Engineer Toolkit (SET) [---]
[---] Created by: David Kennedy (ReLiK) [---]
[---] Version: 8.0.3 [---]
[---] Codename: 'Maverick' [---]
[---] Follow us on Twitter: @TrustedSec [---]
[---] Follow me on Twitter: @HackingDave [---]
[---] Homepage: https://www.trustedsec.com [---]
Welcome to the Social-Engineer Toolkit (SET).
The one stop shop for all of your SE needs.

The Social-Engineer Toolkit is a product of TrustedSec.
```

LAB 2: Website Cloning + Credential Harvesting



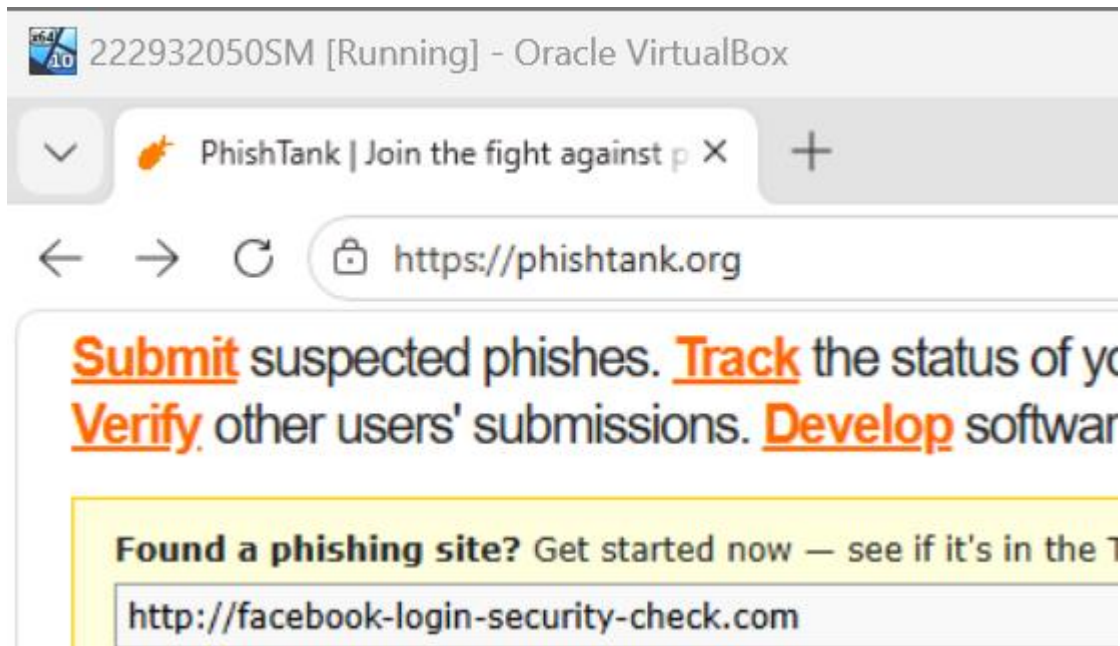
LAB 3: PHISHING EMAIL ANALYSIS (HEADER + URL ANALYSIS)

Field	Value
Sender domain	faceb00k-security.com
Reply domain	evil-domain.ru
Link domain	facebook -login- security-check.com

Component	Meaning
http	Not secure
domain	Fake
/login.php	Credential capture script

LAB 4: NETCRAFT (REAL-TIME DOMAIN RECON + REPUTATION)

LAB 5: PHISHTANK (THREAT INTELLIGENCE VALIDATION)



LAB 6: SOCIAL MEDIA IMPERSONATION (OSINT + ATTACK DESIGN)

Target Name: John Smith

Job: IT Admin

Company: TUT

Email: john@tut.ac.za

Hi, IT support here. We need to verify your account.
Please send your login details.



LAB 7: IDENTITY THEFT (ATTACK CHAIN MODELLING)

OSINT(web-browser) → Data Collection (emaillogin and passwords)→ Identity Creation(fake company) → Exploitation

LAB 8: PHISHING SIMULATION (ENTERPRISE PERSPECTIVE)

Email → User Click → Credential Capture → Report

LAB 9: CORPORATE SOCIAL ENGINEERING RISK

Layer	Risk
User	Trust
System	Weaker authentication
Network	Data leakage